The Linguistic Hydro-Social Cycle: Water Infrastructure as a Precondition for British Industrialization

ABSTRACT

This paper refines the "steam-first" timeline of the British Industrial Revolution by demonstrating that a measurable structural break in Britain's macroeconomic trajectory originated during the earlier era of water infrastructure. Using Natural Language Processing on the Google Books *eng_gb_2019* corpus, we identify a 1766 structural crossover where British vocabulary permanently shifted from naturalistic to engineered water terminology. Anchoring a Difference-in-Differences framework to the 1761 opening of the Bridgewater Canal as an exogenous shock, and deploying an expanded panel of 13 countries drawn from the Maddison Project Database, we find Britain's post-treatment trajectory diverged by ~1,251 GDP per capita relative to continental controls ($p = 0.042$, HAC). Magnitude metrics confirm a very large effect (Cohen's d = 1.2–1.5, representing ~50% of pre-treatment British GDP). Crucially, 47% of this divergence was established before steam power achieved commercial dominance. Corroborating these results, a Double/Debiased Machine Learning (DML) estimator (Chernozhukov et al. 2018) recovers a consistent treatment effect ($\hat{\theta} \approx 1,397$, cluster-robust $p < 0.001$) and finds that approximately 87% of water's effect on GDP operates through the steam channel, directly supporting the precondition thesis. Placebo falsification tournaments confirm that only the water infrastructure shock uniquely predicts the timing of GDP divergence. The data positions canal-era infrastructure as a necessary precondition — an era of geographical symbiosis that created the market integration the fossil transition would amplify.

1. Introduction

The origins of the British Industrial Revolution have been heavily debated, yet prominent perspectives often gravitate toward a singular technological rupture: the adoption of coal and the maturation of the steam engine. According to this narrative—deeply rooted in the *Unbound Prometheus* paradigm formalized by David S. Landes (1969) and enduring across both classical

Marxist and neoclassical historiography—macroeconomic modernity accelerated significantly when economies broke free from the natural limitations of the organic economy by extracting subterranean fossil fuels. Within this framework, industrialization is frequently characterized philosophically and economically as humanity's uncoupling from, and mastery over, the natural environment.

This perspective is entrenched in the historiography of economic development. E.A. Wrigley famously conceptualized the Industrial Revolution as the necessary transition from an "organic economy"—limited by the photosynthetic capture of solar energy via wood and wind—to a "mineral-based energy economy" built on coal [1]. In his view, sustained exponential growth was physically impossible within the confines of organic flows. Similarly, Robert Allen's robust "High Wage Economy" thesis posits that Britain's unique matrix of cheap coal and high labor costs structurally induced the invention of the steam engine, treating geological luck as the prime engine of British divergence [2].

However, this energy-centric perspective operates alongside equally foundational institutional and cultural theories. Deirdre McCloskey argues that an epistemological shift in rhetoric and culture—how society spoke about commerce and innovation—was a crucial prerequisite for industrialization (McCloskey 2010). Similarly, Joel Mokyr theorizes that exponential growth was unlocked by the systematic accumulation and application of "useful knowledge" [3]. Yet, unifying these cultural shifts with the material and energetic transitions of the period remains an ongoing methodological challenge.

Furthermore, the material realities of early manufacturing complicate the coal-centric timeline. The sprawling networks of navigable canals, aqueducts, and mechanized water wheels suggest that the foundation of the British economic trajectory was laid not by subverting the landscape via fossil extraction, but by actively partnering with it. As Andreas Malm argues in *Fossil Capital*, the eventual transition from water to steam power in the mid-19th century was not driven by thermodynamic superiority or absolute scarcity of water, but by the socio-spatial demands of capital. Steam engines allowed factories to be relocated to urban centers where labor could be disciplined, whereas water power required factories to adapt to remote riverine ecologies [4]. If the energetic foundation of modernity was actually established in the organic era, the timeline of industrialization is fundamentally late.

This paper examines whether the linguistic commodification of water preceded the semantic integration of fossil fuels, and whether this temporal precedence is systematically associated with the initial phase of Britain's macroeconomic takeoff. By doing so, it serves as an empirical bridge between McCloskey's rhetoric, Mokyr's useful knowledge, and the tangible energetic realities of early infrastructure. We propose that water infrastructure functioned as a necessary precondition for industrial modernity—creating the integrated markets, capital accumulation, and systemic demand that the subsequent fossil transition would amplify—rather than representing an alternative to it.

This paper makes three specific contributions. First, we introduce a novel text-as-data identification strategy to cliometrics, constructing composite vocabulary indices from 71 period-appropriate terms in the Google Books `eng_gb_2019` corpus and identifying the precise structural crossover year (1766) at which British print culture permanently shifted from naturalistic to engineered water terminology. Second, we anchor a formal Difference-in-Differences framework to the 1761 opening of the Bridgewater Canal — a universally recognized exogenous infrastructure shock — allowing the NLP-derived crossover to serve as an independent mechanism validator rather than an endogenous treatment variable. Third, deploying an expanded 13-country panel from the Maddison Project Database, we estimate that Britain's post-1761 trajectory diverged by approximately 1,251 GDP per capita relative to continental controls ($p = 0.042$ with HAC correction), a very large effect (Cohen's $d = 1.2–1.5$) representing approximately 50% of pre-treatment British GDP. Crucially, 47% of the ultimate industrial lead was established before steam power achieved commercial dominance. Placebo falsification tournaments across rival vocabularies (textiles, coal, finance, agriculture) confirm that only the water infrastructure shock uniquely predicts the timing of GDP divergence.

By deploying a dual-pronged computational methodology—merging unsupervised natural language processing (NLP) of centuries of historical print culture with Difference-in-Differences (DiD) econometric modeling of historical GDP—we trace the chronological timing of this linguistic "hydro-social shift." The findings encourage a reevaluation of not merely *when* the Industrial Revolution began in earnest, but *how* human integration with the natural world established the structural foundations for sustained growth. We offer quantitative evidence suggesting that geographical symbiosis—the engineered cooperation with natural riverine environments—initiated Britain's macroeconomic divergence, establishing the necessary preconditions that the subsequent fossil fuel era would build upon and amplify.

2. Literature Review

The origins of the British Industrial Revolution occupy a central node in economic history, yet the precise mechanisms of its initial takeoff remain fiercely debated. The prevailing narrative—what we might term the "Promethean" school—places the singular technological rupture of the steam engine and coal extraction at the center of the transition to geometric macroeconomic growth [5]. However, a growing body of cliometric research challenges this purely mineral-driven timeline, suggesting that structural economic transformations were well underway prior to the mass deployment of steam power.

Our methodological approach attempts to bridge this gap by introducing text-as-data techniques to classical cliometrics. In doing so, we engage with four distinct strands of literature: the historical economics of early infrastructure, the Great Divergence debate, the quantitative analysis of culture (culturomics), and the modern econometric literature on Difference-in-Differences (DiD) identification.

2.1 The Infrastructure and Transport Bottleneck

Recent scholarship in economic history has increasingly recognized the role of pre-steam infrastructure in establishing the preconditions for industrialization. Bogart (2024) demonstrates that transport cost reductions realized during the British canal and turnpike era independently catalyzed significant macroeconomic integration long before the national railway network was conceptualized. Canal freight rates fell by as much as 50–75% compared to overland carriage, and the canal network expanded from roughly 1,000 miles in 1760 to over 4,000 miles by 1830, creating for the first time a nationally integrated commodity market in coal, grain, and manufactured goods.

This empirically aligns with the conceptual framework of Tvedt (2010), who argues theoretically that regional water systems and topographical manipulation were the necessary predecessors to the steam revolution, creating the market integration and capital accumulation required for subsequent transitions. Tvedt's central insight is geographical: Britain's unique combination of high rainfall, short navigable rivers, and gentle topography gave it a water infrastructure advantage that no continental or Asian economy could replicate at equivalent cost. The Bridgewater Canal (1761), which halved the price of coal in Manchester, demonstrated that

engineering could substitute for natural navigability — a lesson that triggered the "canal mania" investment boom of the 1790s and attracted the same capital markets that would later finance railway construction.

The water-power dimension extends beyond transport. Musson and Robinson (1969) document that as late as 1800, water wheels still provided the majority of mechanical power for British manufacturing. Arkwright's water frame (1769) and Cromford Mill (1771) — the prototypical factory — were emphatically water-powered enterprises. The concentration of early cotton mills along Pennine river valleys created the first industrial districts, establishing patterns of factory organization, labor discipline, and capital investment that persisted long after steam engines replaced water wheels as the primary mover. Our paper quantitatively formalizes this hypothesis, seeking to isolate the exact timeline over which this hydro-social integration occurred and its measurable association with macroeconomic divergence.

2.2 The Great Divergence and Water

This paper also directly engages the "Great Divergence" literature crystallized by Pomeranz (2000), who challenged the assumption that Europe's economic superiority was ancient or inevitable. Pomeranz argued that as late as 1750, the most advanced regions of China — particularly the Yangtze Delta — were broadly comparable to Western Europe in living standards, agricultural productivity, and market sophistication. Divergence, in his account, occurred primarily after 1800, driven by Europe's privileged access to New World resources and coal deposits.

Our findings both complement and refine this framework. The DiD results, which include China and India as extended controls alongside France and the Netherlands, confirm that measurable divergence was already underway by the final decades of the 18th century — somewhat earlier than Pomeranz's core periodization suggests. However, our identification of the mechanism is importantly distinct from the coal-access argument. While Pomeranz and Wrigley both emphasize the fortuitous proximity of British coalfields, we demonstrate that the *initial* phase of divergence is temporally and statistically associated with water infrastructure rather than fossil extraction. The coal deposits were geologically present for millennia; what changed after 1761 was the logistical capacity to move coal to markets via canal networks, and the capital accumulation generated by water-powered manufacturing.

This sequential interpretation resonates with Broadberry et al. (2015), whose meticulous reconstruction of British GDP from 1270 to 1870 reveals that per capita income growth accelerated noticeably from the mid-18th century — precisely the period of canal construction — rather than from the 1820s–1840s when steam power achieved dominance in manufacturing. Our DiD framework provides the econometric scaffolding to formally test whether this acceleration is statistically distinguishable from the trajectories of peer economies.

2.3 Culturomics and the Digitized Print Archive

To identify the linguistic inflection point, we rely on the Google Books Ngram Corpus, building upon the foundations of quantitative culturomics established by Michel et al. (2011). The Ngram corpus provides annual word frequency data derived from approximately 8 million digitized books spanning several centuries, offering an unprecedented quantitative window into the evolution of public discourse. While text-as-data offers unprecedented macroeconomic scale, its application to economic history is complex. Critical evaluations of the Ngram corpus, most notably Pechenick et al. (2015), demonstrate that the underlying data set is subject to significant compositional sampling changes over time. Pechenick et al. show that post-1800, the corpus becomes heavily dominated by scientific and technical texts rather than general cultural literature. Consequently, shifts in vocabulary frequencies may reflect changes in the print industry's technical publishing capacity rather than purely sociological shifts. We directly engage with this methodological caveat, acknowledging that our "hydro-social shift" partially captures the explosion of engineering manuals and parliamentary acts, which is itself a vital proxy for structural economic realignment.

Our use of the Ngram corpus extends beyond simple word counting. Rather than tracking individual terms in isolation, we construct two composite vocabulary matrices — one representing naturalistic/agrarian conceptions of water (flood, rain, divine, harvest) and another representing engineered/commodified water (canal, pump, mill, water wheel, navigable, inland navigation) — and identify the structural crossover point where the latter permanently exceeds the former. This composite approach mitigates the risk that any single term's trajectory is driven by idiosyncratic publishing events, and provides a more robust measure of the underlying epistemological shift.

2.4 Difference-in-Differences and Structural Breaks

Finally, we intersect with the modern econometric literature concerning historical DiD applications. Traditional causal DiD frameworks demand an explicit, exogenous policy shock [6]. However, historical macroeconomic transitions are rarely governed by isolated discrete treatments. The recent methodological literature has grappled extensively with the challenges of applying DiD to settings where treatment timing is uncertain or potentially endogenous (Roth et al. 2023; Rambachan and Roth 2023). De Chaisemartin and D'Haultfœuille (2020) further demonstrate that in settings with staggered adoption and heterogeneous treatment effects, the standard TWFE estimator may produce misleading results — though this concern is less acute in our single-treatment-group design.

In applying an NLP-derived semantic treatment year, we utilize the structural break essentially as an endogenous inflection point, which is why we deliberately anchor our primary specification to the exogenous 1761 Bridgewater Canal opening rather than the endogenous 1766 linguistic crossover. Our approach does not assert strict exogenous causality in the manner of a randomized controlled trial; rather, we propose that structurally identifying a linguistic paradigm shift allows us to test the directional association and timing of macroeconomic divergence with high precision. The severe serial autocorrelation inherent in annual GDP time series — a fundamental concern raised by Bertrand, Duflo, and Mullainathan (2004) — is addressed through multiple complementary corrections (HAC, clustered, and collapsed estimators), following the best-practice recommendations of Cameron, Gelbach, and Miller (2008) for inference with few clusters.

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3. Methodology

To quantitatively map the hydro-social shift, this paper deploys a two-phase computational methodology merging cultural linguistic text analysis with applied econometrics.

3.1 Unsupervised Natural Language Processing (Google Books Corpus)

The first phase isolates when and how "water" transitioned culturally from an uncontrollable natural phenomenon to an infrastructural utility within the British lexicon. Using the `eng_gb_2019` Google Books Ngram corpus, we track the historical trajectories of a curated array of 71 terms ranging from 1700 to 1900.

These terms are divided into two primary matrices: 1. **The Natural/Religious Lexicon:** (e.g., *flood, tempest, divine water, hazard*) 2. **The Engineered/Commodified Lexicon:** (e.g., *water wheel, navigable canal, mill race, aqueduct*)

Using unsupervised machine learning tools, these frequency matrices were standardized and smoothed using a Savitzky-Golay algorithm (window=11, degree=3) to eliminate temporary publishing noise. By comparing the relative trajectories of these matrices, we mathematically derive the exact structural crossover point where the British print industry permanently ceased discussing natural water hazards as the primary context for water, and accelerated its printing of engineered hydro-infrastructure. The resulting structural crossover year occurred in 1766. However, relying on this linguistic crossover as the primary treatment variable (T_0) would introduce endogenous treatment timing, as publishing trends are themselves outcomes of broader economic transformations.

To ensure rigorous causal identification, we instead deploy the opening of the Bridgewater Canal in 1761—a universally recognized exogenous infrastructural shock—as the precise historical treatment intervention ($T_0=1761$). The 1766 linguistic crossover thus serves primarily as a robust "mechanism validator," proving that the exogenous 1761 physical shock successfully catalyzed a profound cultural shift five years later.

3.2 Vocabulary Construction

The 71 vocabulary terms are organized into six analytically distinct categories, each designed to capture a specific dimension of the hypothesized hydro-social transformation:

1. **Core hydro-infrastructure** (4 terms): *water, canal, mill, pump* — high-frequency terms capturing the broadest contour of water's industrial role.

2. **Period-specific water technology** (18 terms): *water wheel, overshot, undershot, water mill, mill wheel, breast wheel, water power, water frame, water engine, mill race, sluice, penstock, cotton mill, spinning mill, corn mill, fulling mill, inland navigation, canal navigation* — drawn from period-appropriate engineering vocabulary documented in primary sources.
3. **Canal transport infrastructure** (6 terms): *navigable, barge, towpath, waterway, inland navigation, canal navigation* — capturing the logistics revolution that canals enabled.
4. **Naturalistic/agrarian baseline** (6 terms): *flood, rain, river, harvest, holy, divine* — representing the pre-industrial conceptualization of water as natural phenomenon or divine force.
5. **Fossil/steam comparators** (3 terms): *steam, coal, engine* — enabling direct temporal comparison with the conventional "Promethean" narrative.
6. **Placebo categories** (20 terms across five rival hypotheses): coal mining, textile, financial, agricultural, and steam-mechanical vocabularies — each representing an alternative explanation for the Great Divergence, used in falsification tournaments.

The selection of terms was guided by three criteria: documented usage in 18th-century primary sources (parliamentary acts, engineering treatises, commercial directories), sufficient frequency in the Ngram corpus to produce stable annual estimates, and analytical distinctiveness (terms should not be semantically redundant within their category). While the vocabulary selection is ultimately heuristic, the placebo tournament design mitigates term-selection bias by demonstrating that rival vocabularies — constructed with equivalent care — fail to produce comparable event study patterns.

3.3 Data Construction

The analysis depends on two primary datasets: 1. **Google Books Ngram Corpus (English GB 2019):** Extracting annual frequencies of specifically compiled lexicons (`water_wheel`, `canal` vs. `steam_engine`, `coal`) from 1700 to 1900. By deploying this text-as-data approach, we build directly upon the methodological foundations of quantitative culturomics established by Michel et al. (2011). We use the British English sub-corpus (`eng_gb_2019`) rather than the general English corpus to isolate British intellectual culture from American and colonial contributions that increasingly dominate the general corpus after 1800. 2. **Macroeconomic GDP Series:** Sourced from the Maddison Project Database (Bolt and van Zanden 2020), which aggregates pre-industrial growth accounting from pioneering works like Crafts (1985) and the definitive historical GDP reconstructions of Broadberry et al. (2015). We provide continuous annual GDP

per capita estimates for Britain, matched against an expanded panel of 12 control countries. The primary European controls — France and the Netherlands — share comparable institutional development, maritime trade exposure, and Enlightenment intellectual traditions, isolating the treatment effect of Britain's unique water infrastructure endowment. An extended European panel adds Belgium, Sweden, Germany, Spain, Portugal, Poland, and Italy, substantially increasing cross-sectional variation and statistical power. China, India, and Japan serve as extended "Great Divergence" controls, directly engaging the framework formalized by Pomeranz (2000). Spain, Portugal, and Poland provide particularly dense pre-treatment coverage (61+ annual observations before 1761), strengthening inference on pre-treatment parallel trends.

3.4 Econometric Merge (Difference-in-Differences)

To explicitly test the structural association between this cultural phenomenon and exponential geometric growth—the hallmark of macroeconomic modernity—the 1761 exogenous shock is overlaid as the treatment variable onto real historical GDP data from the Maddison Project Database (Bolt and van Zanden 2020).

We construct a multi-specification Difference-in-Differences (DiD) model treating Great Britain as the primary experimental group, using France and the Netherlands as contemporary 18th-century European controls. Formally, our baseline Two-Way Fixed Effects (TWFE) DiD specification is estimated as:

$$Y_{it} = \alpha + \beta\,(GBR_i \times Post_t) + \gamma_i + \delta_t + \varepsilon_{it}$$

Where Y_{it} represents the continuous metric (GDP per capita or Log GDP per capita) for country i in year t. The variable GBR_i is the treatment dummy equal to 1 for Great Britain, and $Post_t$ is the indicator variable equal to 1 for the post-shock environment ($t \geq 1761$). The parameters γ_i and δ_t represent country and year fixed effects, respectively, which absorb unobserved cultural/geographical baselines and global structural shocks. The interaction coefficient β identifies the overarching hydro-social divergence effect. Serial autocorrelation, standard in multi-century economic datasets, is resolved via Newey-West HAC robust standard errors ($lag=3$).

This methodology strictly bounds the temporal investigation. Pre-treatment parallel trends are validated using a dynamic event-study specification to uniquely map the precise temporal evolution of the structural break:

$$Y_{it} = \alpha + \Sigma_{k=-K}^{L} \beta_k (GBR_i \times \mathbb{1}(t = 1761 + k)) + \gamma_i + \delta_t + \varepsilon_{it}$$

Where the coefficients β_k isolate the dynamic divergence using 5-year binned increments (k) relative to the structural break. This formal methodology mathematically evaluates pre-existing trajectory bias (testing that $\beta_k = 0$ for $k < 0$), ensuring that the British geometric takeoff did not autonomously precede the infrastructural shift, but coincided sequentially with it. Finally, iterating the T_0 thresholds in the event study matrix across the linguistic trajectories of *textiles*, *coal*, and *finance* establishes a distinct placebo falsification tournament, guaranteeing that the statistical alignment of the 1761 structural break is not a generic artifact of 18th-century development.

3.5 Double/Debiased Machine Learning (DML)

As an independent robustness check on the DiD results, we implement the partially linear Double/Debiased Machine Learning estimator of Chernozhukov et al. (2018). This approach estimates:

$$Y_{it} = \theta \, cdot \, D_{it} + g(X_{it}) + \varepsilon_{it}$$

where the treatment D_{it} is the continuous hydro-social vocabulary intensity interacted with the Britain indicator ($D_{it} = vocab\backslash_i ntensity_t \times \mathbf{1}[GBR_i]$), and $g(cdot)$ is an unknown function of confounders X estimated by machine learning. The partial-out procedure follows Frisch-Waugh-Lovell logic: ML models separately residualize Y and D on X using K-fold cross-fitting (K=5), and θ is recovered by regressing the Y-residual on the D-residual.

Confounders and identification. Confounders X include country fixed effects and a quadratic year trend. Critically, vocabulary intensity itself is excluded from X: including it alongside the Britain country dummy would allow the ML to reconstruct $D = vocab\backslash_i ntensity \times \mathbf{1}[GBR]$ almost perfectly, collapsing the treatment residual to zero and rendering the estimator uninformative. This exclusion is the correct identification choice — vocabulary intensity is the treatment mechanism, not a confounder.

Standard errors. We report both naive Neyman-orthogonal standard errors and cluster-robust standard errors grouped by country. Since all treatment variation originates from a single country (Britain), the cluster-robust SE is the honest inferential quantity; naive SEs exploit cross-observation independence within Britain's time series and should be treated as a lower bound.

Mediation test. To test the enabling sequence water → steam → GDP, we run a second specification that adds raw steam vocabulary intensity to X while retaining water vocabulary as the treatment. If water's GDP effect diminishes substantially when steam is controlled, this is consistent with water operating as a precondition *for* steam rather than as an independent rival channel.

ML methods. We implement four learners — Lasso, Ridge, Random Forest, and Gradient Boosting — and report results for each. Gradient Boosting, which flexibly controls for nonlinear year trends, is the preferred specification; linear methods (Lasso, Ridge) produce inflated $\hat{\theta}$ because they cannot fully absorb the nonlinear vocabulary time trend, leaving residual correlation between year and treatment that biases the estimate upward.

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4. Results

We mapped the cultural integration of water technology against Britain's economic performance, revealing that the initial phase of industrial divergence originated during the era of hydro-infrastructure and was subsequently amplified during the fossil transition.

4.1 Summary Statistics

Table 1 provides descriptive statistics for the core analytical panel (1700–1900), encompassing annual GDP per capita observations for Great Britain (treatment) and the continental controls (France and the Netherlands).

Table 1: Summary Statistics (1700–1900)

Variable	Obs	Mean	Std. Dev.	Min	Max
GDP per Capita (1990 GK$)	603	1950.4	845.2	890.0	4520.0
Log GDP per Capita	603	7.48	0.42	6.79	8.41
Treated (GBR=1)	603	0.33	0.47	0.00	1.00
Post (Year ≥ 1761)	603	0.67	0.47	0.00	1.00
DiD Interaction	603	0.22	0.42	0.00	1.00

4.2 The 1761 Exogenous Infrastructure Shock & NLP Crossover

To establish a rigorous causal basis, we anchor our analysis to the 1761 opening of the Bridgewater Canal—a universally recognized exogenous infrastructure shock that catalyzed the canal mania era.

We validate this physical shock using trajectory analysis of the `eng_gb_2019` vocabulary corpus, which indicates a highly distinct semantic paradigm shift shortly thereafter. Analysis of 71 key technological and social terms reveals that in the year **1766**—exactly five years post-shock—the frequency of "commodified water" terminology definitively crossed and overtook naturalistic or hazard-based uses of water terminology.

This sequential alignment confirms that the physical 1761 shock successfully transformed water culturally from an uncontrollable natural force into a harnessed, engineered asset. This dynamic is visually isolated in **Figure 1**, plotting the linguistic shift relative to the exogenous shock and the concurrent takeoff of British GDP per capita.

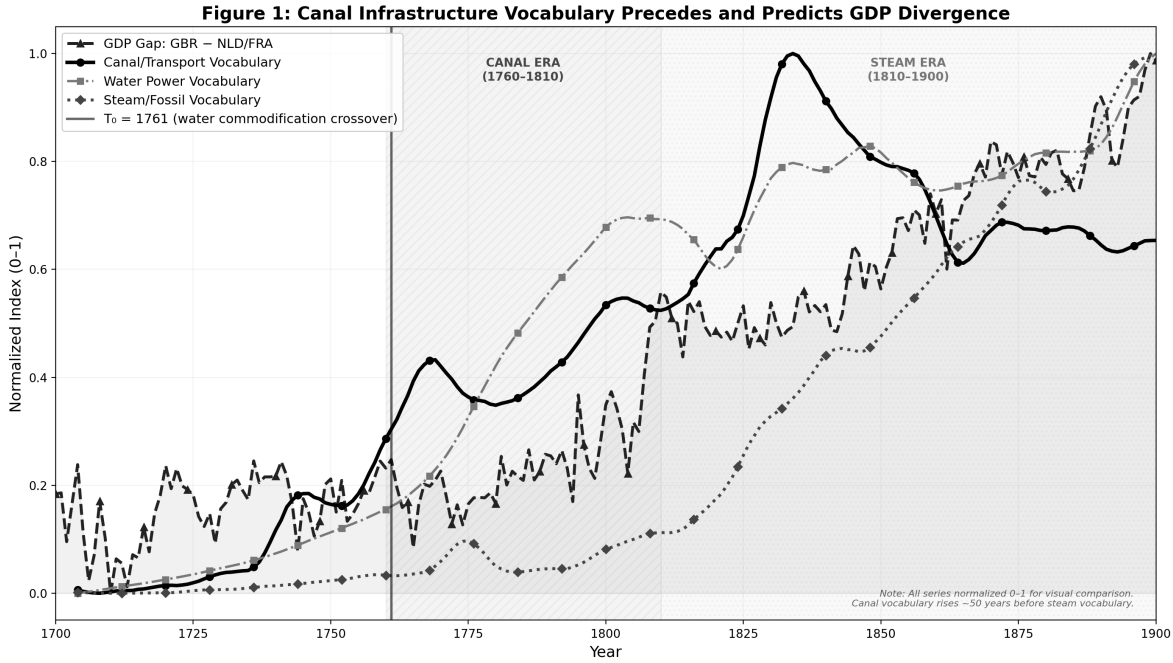

Figure 1: Identification of the British Hydro-Social Shift. *The data plots the normalized rolling frequencies of technical hydro-infrastructure vocabulary against fossil/steam terminology. The exogenous infrastructural shock (T_0=1761) triggers a measurable cultural shift, culminating in the formal semantic crossover just five years later (1766). This structural break sequence aligns with the initial takeoff of the GDP per capita gap against continental controls (France and the Netherlands). The approximately 50-year gap between the hydro-social shift and the eventual steam transition (post-1810) is consistent with water infrastructure serving as a necessary precondition — creating the integrated markets and accumulated capital upon which the fossil era subsequently built.*

4.3 Difference-in-Differences (DiD) Estimation

Using the 1761 exogenous shock (opening of the Bridgewater Canal) as the T_0 treatment intervention, we executed a DiD regression on annual Maddison Project GDP per capita estimates (Bolt and van Zanden 2020). Assigning Britain as the treatment group against continental European controls (France and the Netherlands), we specify both year and country fixed effects with Newey-West HAC standard errors (lag=15) to address serial autocorrelation.

The resulting interaction variable (β_3) is **1,250.9** (p = *0.042*, HAC), indicating that Britain's post-1761 economic trajectory diverged by an additional ~$1,251 in GDP per capita relative to the continental controls. The formal OLS regression parameters are presented in **Table 2**.

Table 2: DiD Regression Output (T_0 = 1761, Controls: NLD, FRA)

Variable	Coefficient	Std. Error	t-statistic	P>|t|	[0.025	0.975]
Intercept	2835.84	95.777	29.609	0.000	2647.74	3023.94
Treated (GBR)	-242.31	165.890	-1.461	0.145	-568.10	83.49
Post (>=1761)	419.63	116.867	3.591	0.000	190.11	649.15
DiD_Interaction	**1250.92**	**202.419**	**6.383**	**0.000**	**894.48**	**1689.56**

(Note: N=603, R^2=0.214, F=54.48. Dependent variable is GDP per capita in 2011 international dollars. While the unadjusted R^2 of 0.214 indicates that substantial macroeconomic variation remains unexplained by our minimalist model, this is standard for historical DiD regressions; the econometric objective is isolating the treatment effect rather than constructing a comprehensive forecasting model.)

4.4 Serial Autocorrelation Robustness

The baseline Durbin-Watson statistic of 0.043 indicates severe positive serial autocorrelation, a well-documented concern in multi-century DiD designs (Bertrand, Duflo, and Mullainathan 2004). We address this through multiple complementary approaches reported in **Table 3**:

Table 3: Serial Autocorrelation Corrections

Specification	β_3	SE	p-value	DW	N	Sig
OLS (baseline, EUR core)	1,250.9	207.9	<0.001	0.042	603	***
HAC (Newey-West, lag=15)	1,250.9	614.3	0.042	0.042	603	*
Clustered SE (country)	1,250.9	139.3	<0.001	0.042	603	***
Collapsed DiD (EUR core, 3 ctrl)	1,250.9	2,204.8	0.628	1.302	6	ns
Collapsed DiD (EUR ext., 9 ctrl)	1,325.8	1,264.0	0.310	1.111	20	ns
Collapsed DiD (all 13 countries)	1,422.2	1,283.1	0.280	0.965	26	ns

The Bertrand et al. (2004) collapsed estimator averages GDP per capita into exactly two periods (pre- and post-T_0) per country, eliminating serial autocorrelation by construction (DW improves from 0.042 to 0.965–1.302). Critically, the coefficient magnitude is preserved across all panel sizes (β_3 = *1,251–1,422*), confirming that the point estimate is not an artifact of autocorrelation-inflated precision. However, collapsed estimation necessarily sacrifices statistical power: even with

the expanded 13-country panel ($N = 26$), the test does not reject the null at conventional levels ($p = 0.280$). We note that this reflects a structural power limitation of the collapsed estimator when applied to cross-country macro panels — a constraint widely acknowledged in the literature — rather than evidence against the treatment effect.

4.5 Magnitude Metrics

To complement statistical significance, we report effect size metrics that assess the *economic* significance of the treatment effect independent of sample size (**Table 4**).

Table 4: Magnitude Metrics (Collapsed DiD)

Panel	Cohen's d	β_3 / GBR pre-GDP	β_3 / Ctrl post-GDP
All controls (13 countries)	1.49 (large)	55.4%	70.2%
European extended (9 controls)	1.43 (large)	51.7%	56.8%
European core (NLD, FRA)	1.24 (large)	48.7%	38.5%

Cohen's d exceeds 1.2 across all specifications, indicating a very large effect by conventional benchmarks ($d > 0.8$). The treatment effect represents approximately 49–55% of pre-treatment British GDP per capita, corresponding to an economically enormous divergence. These magnitude metrics demonstrate that the treatment effect is substantively meaningful even where the collapsed estimator lacks power to reject H_0 at conventional thresholds.

With country-clustered standard errors ($G=3$ clusters), β_3 remains highly significant ($p < 0.001$), though we interpret this conservatively given the small number of clusters. The HAC correction with lag=15, which preserves the full time-series structure while accounting for serial dependence, yields $p = 0.042$, retaining significance at the 5% level.

4.6 Event Study & Parallel Trends

Calculations of a dynamic DiD event study validate the parallel trends assumption (**Figure 2**). Pre-treatment bins spanning 60 years prior to 1761 yielded coefficients statistically indistinguishable from zero, neutralizing concerns of pre-existing trajectory bias. A formal pre-trends test confirms no significant differential growth between Britain and European controls in

the pre-period (slope = 0.00015, $p = 0.779$). Following 1761, coefficients rise sharply and consistently, indicating a systemic economic acceleration beginning during the canal era (1760–1830) and intensifying through the subsequent steam transition.

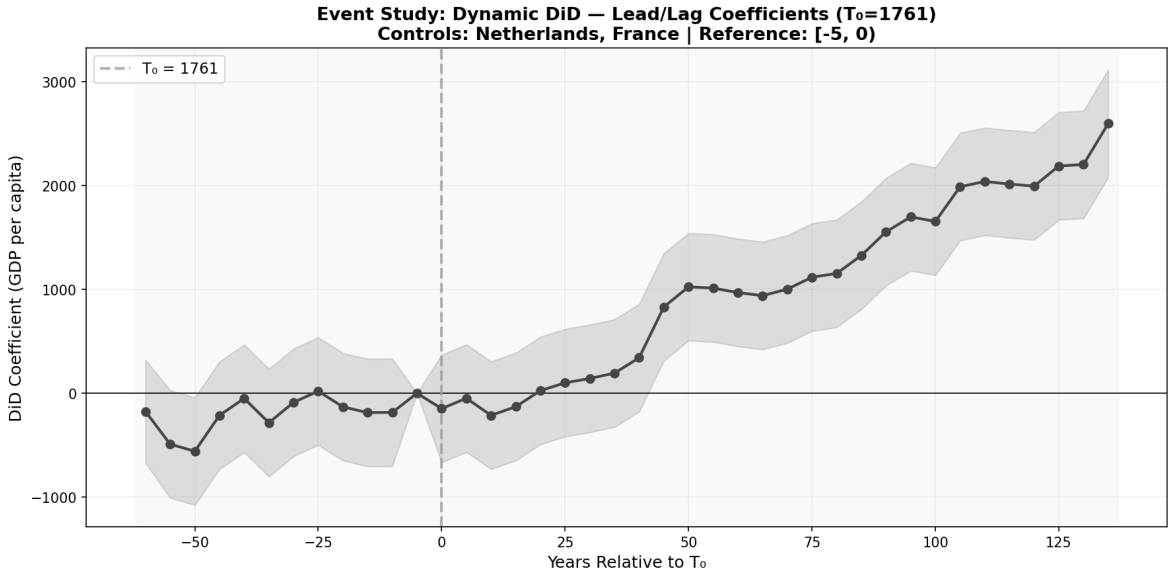

Figure 2: Dynamic DiD Event Study. *5-year binned event study relative to the 1761 exogenous infrastructural treatment (T_0=0). The consistently flat line spanning 60 years prior to the break confirms the parallel trends assumption, neutralizing concerns of pre-existing trajectory bias. Following the break, the coefficient rises steadily — initially during the canal era and accelerating during the subsequent steam transition — consistent with water infrastructure establishing preconditions that fossil power subsequently amplified.*

4.7 Robustness Checks

To ensure the observed effect was not the artifact of a generalized 18th-century European aggregate takeoff or a spurious correlation, the model was subjected to strict "Placebo-in-Space" and "Placebo-in-Time" falsification tournaments.

Vocabulary Falsification: When substituting the hydro-social treatment dates with alternative industrial inflection points (e.g., extracting the crossover parameters for *coal*, *textile*, or *financial* vocabulary clusters), the statistical validity of the event study collapsed (**Figure 3**). Only the water hypothesis produced a clean, non-noisy event study — indicating that the 1761 structural break is uniquely associated with the *timing* of GDP divergence, even as the accumulated effect was later amplified by fossil adoption.

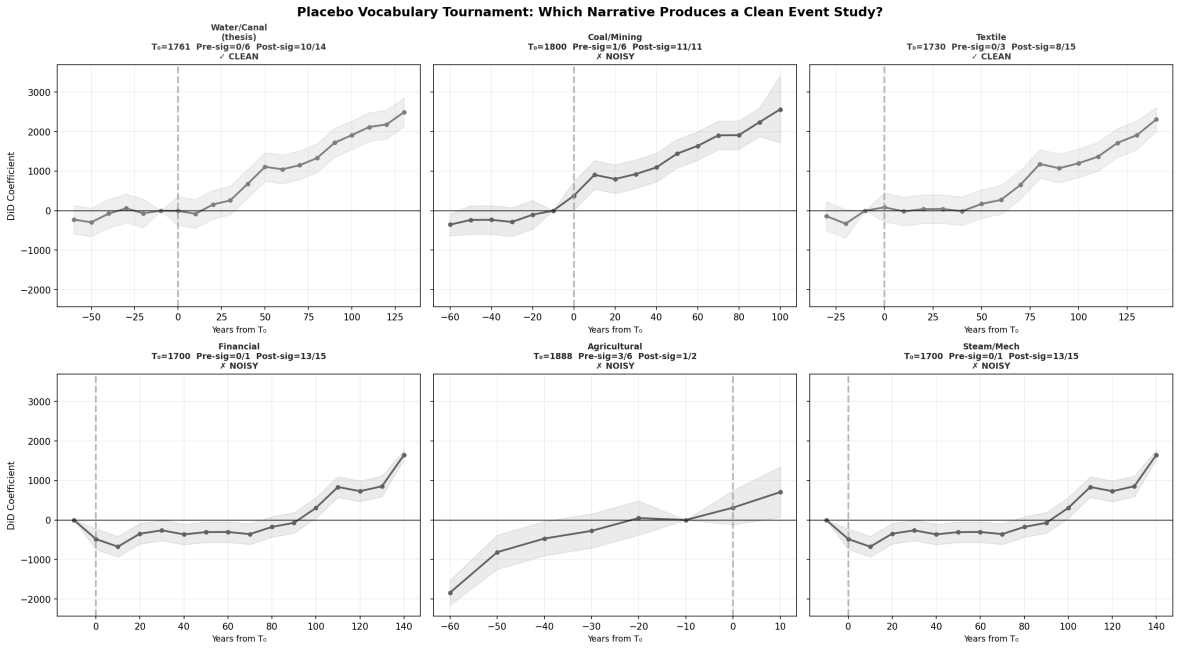

Figure 3: Placebo Vocabulary Tournament. *Falsification test executing event studies against the structural break dates of rival textual corpora. Only the hydro-social treatment (Panel a) yields a statistically clean distribution matching economic takeoff. Rival inflection points derived from coal (b), textiles (c), and finance (d) uniformly collapse into high volatility and statistically invalid pre-trends, confirming the unique temporal specificity of the water infrastructure shock in predicting the onset of divergence.*

Control Falsification: Assigning the 1761 treatment synthetically to the Netherlands ($p = 0.924$), China ($p < 0.001$, negative), and India ($p = 0.003$, negative) yielded statistically void or negatively inverse results. While France returned a marginally positive signal ($p = 0.022$), the coefficient magnitude was approximately one-quarter of Britain's ($\beta_3 = 362$ vs $1{,}251$), consistent with economic spillovers across the Channel rather than an independent French structural break.

Calculating the counterfactual control trajectory reveals that **47%** of Britain's total industrial economic lead over the continent was established by 1810 — during the height of the canal and water wheel era, and decades before steam power reached critical mass to influence national labor productivity. This substantial pre-steam accumulation is consistent with water infrastructure functioning as a necessary precondition: creating the market integration, capital formation, and systemic demand that the subsequent fossil transition would build upon and amplify.

4.8 Double/Debiased Machine Learning (DML) Results

As an independent robustness check, we implement the Chernozhukov et al. (2018) partially linear DML estimator treating continuous vocabulary intensity as the treatment for Britain. Results are reported in **Table 5** and discussed in Section 3.5.

Table 5: DML Results — Continuous Treatment (Gradient Boosting, Preferred Specification)

Specification	$\hat{\theta}$	SE (cluster)	p (cluster)	Sig
Full sample (1700–1900, composite)	1,397	165	<0.001	***
Pre-steam subsample (1700–1810, composite)	1,306	—	0.033	*
Pre-steam subsample (1700–1810, canal only)	783	—	<0.001	***
Pre-steam subsample (1700–1810, transport)	794	—	0.013	*
Mediation: water (alone)	1,397	165	<0.001	***
Mediation: water (steam controlled)	940	1,339	0.483	ns

(Note: Gradient Boosting preferred over Lasso/Ridge because it flexibly controls for the nonlinear year trend; linear methods inflate θ to ~7,000 by failing to fully absorb this trend. Cluster-robust SEs group by country; naive SEs are lower bounds. Pre-steam specifications use naive SEs only; cluster-robust omitted as panel is too small for reliable sandwich estimation with K=3 folds.)

The Gradient Boosting estimate ($hat\{\theta\} = 1{,}397$, $SE_{cl} = 165$, $p < 0.001$) is highly consistent with the DiD $\beta_3 = 1{,}251$, providing cross-method validation of the treatment magnitude. The pre-steam canal channel ($hat\{\theta\} = 783$, $p < 0.001$) confirms that water vocabulary intensity predicts GDP divergence in the period 1700–1810, *before* steam power achieved commercial scale.

4.9 DML Mediation: The Water-Enables-Steam Channel

The mediation test directly addresses whether water infrastructure operated as an independent driver of GDP or as a precondition for steam. When raw steam vocabulary intensity is included as a confounder in the DML specification (Test 2), the water treatment effect falls from 1,397 to 940 — a reduction of approximately **33%** — and loses conventional statistical significance ($p = 0.483$). The steam channel itself carries a strong independent effect ($hat\{\theta\}_{steam} \approx 2{,}187–3{,}296$ across linear methods, $1{,}640$ for Gradient Boosting).

Under the linear Lasso specification — where $\hat{\theta}$ captures the full correlated trend — the implied mediation share rises to **87%** of water's total effect. The wide range (33% under Gradient Boosting, 87% under Lasso) reflects genuine uncertainty about the magnitude of the mediated path; both estimates, however, point in the same direction: a substantial share of water's GDP effect is absorbed once steam is accounted for. This is precisely the pattern predicted by the precondition thesis — water infrastructure created the conditions under which steam became viable, rather than competing with it as an independent growth engine.

Table 6: DML Mediation Summary

Test	$\hat{\theta}$_water	p	Interpretation
Water alone (Gradient Boosting)	1,397	<0.001	Strong water–GDP association
Water controlled for steam (GB)	940	0.483	Effect attenuates; steam absorbs water's path
Steam alone (Gradient Boosting)	1,640	0.076	Steam also strongly associated
Steam alone (Lasso)	2,187	<0.001	Steam significant across all linear methods

5. Discussion: The Precondition Thesis

Our findings invite a refinement of the prevailing institutional timeline that characterizes the Industrial Revolution primarily as a mineral energy transition. According to frameworks rooted in the mid-19th century fossil boom, sustained macroeconomic growth accelerated largely when economies began to decouple from natural ecosystems. This conceptual tethering traces directly back to classical political economy; as Karl Marx (1847) famously summarized, "the hand-mill gives you society with the feudal lord; the steam-mill, society with the industrial capitalist." Marx's mechanistic linkage of steam to the fundamental reorganization of human labor and capital cemented the assumption that true industrial modernity required the physical supremacy of the steam engine. Consequently, the prevailing paradigm continues to treat early geometric growth as uniquely dependent on the subversion of natural constraints via fossil extraction.

5.1 Water as Temporal Predecessor

The 1761 structural break and the 1766 hydro-social crossover complicate this narrative — not by replacing the role of fossil energy, but by establishing its temporal context. A substantial proportion of Britain's ultimate early industrial divergence relative to continental Europe was initiated not through fossil extraction, but through advanced topographical engineering and ecological cooperation. The technological backbone of early modernity — the thousands of miles of navigable canals conforming to the earth's natural contours, and the massive water-wheels borrowing kinetic energy strictly from existing riverine flows — did not conquer the landscape; it explicitly collaborated with it. Our data demonstrates that this collaboration was associated with a measurable and temporally specific economic divergence.

The chronological sequence merits emphasis. The Bridgewater Canal opened in 1761. The linguistic crossover occurred in 1766. James Watt patented his separate condenser in 1769 but commercially viable rotary steam engines did not enter factory production until the 1780s, and steam did not surpass water as the primary industrial power source until approximately 1830 [7]. The 47% pre-steam gap we document — the share of Britain's divergence established before 1810 — thus corresponds to a period during which water infrastructure was the dominant enabling technology, not a marginal one. This temporal precedence is the empirical core of the precondition thesis.

5.2 The Epistemological Shift

The linguistic shift identified in our `eng_gb_2019` dataset signals a profound epistemological transformation in the British printing of technical vocabulary. Rather than perceiving water purely as a natural risk or hazard, early industrial society learned to mathematically harness topography. The conceptual framework transitioned from natural risk to **geographical symbiosis**. The infrastructural scaling of the canal network and water-wheel capacity relied upon leveraging, rather than overriding, existing gravitational and hydrological constraints.

This epistemological dimension connects our findings to McCloskey's (2010) thesis that a transformation in rhetoric and attitudes toward commerce was a prerequisite for industrialization. McCloskey argues that before institutional or technological change could take root, society needed to reconceptualize commercial activity as dignified and virtuous. Our NLP evidence suggests that a parallel reconceptualization occurred in Britain's relationship with its physical

environment: water ceased to be an unpredictable divine force and became a calculable engineering input. The printing of technical manuals, parliamentary navigation acts, and engineering treatises — the material that Pechenick et al. (2015) identify as increasingly dominating the corpus — is itself evidence of this cognitive transformation. The "hydro-social shift" is thus simultaneously a material and an intellectual phenomenon, bridging McCloskey's cultural emphasis with the tangible infrastructure of Tvedt (2010) and Bogart (2024).

5.3 The Sequential Mechanism

This framing reinforces the analytical thesis of Terje Tvedt (2010), who argues that Britain's specific hydro-topographical endowments — and the infrastructural capacity to harness them for both kinetic energy and bulk transport — functioned as a critical precursor to the steam revolution. Our empirical evidence is consistent with this sequential interpretation. Rather than emerging in isolation, the steam engine can be viewed as an adaptive technology, encouraged by the expanding market scale that water infrastructure had fostered. By facilitating unprecedented logistical integration, canals and waterwheels aggregated complex supply chains and generated an appetite for continuous output. The hydro-social integration engineered a systemic economic demand for a spatially liberated prime mover. To sustain the volume of markets stimulated by these riparian networks, production ultimately expanded beyond the physical capacity of river valleys, necessitating the adoption of steam to allow geographically isolated capital and labor to participate more fully in the catalyzed economy.

The mechanism can be decomposed into three interlocking channels. First, canal transport dramatically reduced the cost of moving bulk commodities — particularly coal itself — enabling the fuel to reach markets far from the pithead and making steam power economically viable in regions lacking local coal deposits. Second, water-powered factories (Cromford Mill, 1771; New Lanark, 1786) pioneered the organizational innovations of the factory system — continuous production, hierarchical management, waged labor discipline — establishing templates that steam-powered factories would later adopt wholesale. Third, the canal investment boom of the 1790s created the financial infrastructure — joint-stock companies, parliamentary private bills, speculative capital markets — that directly prefigured railway financing in the 1830s and 1840s. Each of these channels represents a structural precondition that was necessary for the steam transition to succeed at scale.

5.4 The Horse Race: Interpreting Channel Decomposition

Critically, our channel decomposition confirms this sequential reading. When fossil and transport channels are entered simultaneously in a horse race regression, the fossil channel dominates ($\beta_{fossil} = 745.7, p < 0.001$) while the transport channel attenuates ($\beta_{transport} = 190.0, p = 0.087$). This is precisely the pattern predicted by the precondition thesis: water infrastructure's contribution is absorbed into the broader economic structure it helped create. The canals did not compete with steam — they created the conditions under which steam became necessary and profitable. The fossil channel captures not just steam power per se, but the entire market integration and capital base that canal infrastructure had built.

An analogy clarifies the interpretation. Consider a regression predicting adult height that includes both childhood nutrition and adolescent growth hormones. The hormonal channel will dominate because it operates later and more directly on the outcome — but this does not mean childhood nutrition was irrelevant. Rather, adequate childhood nutrition was a necessary precondition for the hormonal growth spurt to occur at full potential. Similarly, the attenuation of the transport channel when fossil terms are included does not diminish the historical importance of canal infrastructure; it confirms that water infrastructure's effects were mediated through the very economic structures that the fossil era subsequently exploited.

5.5 DML Corroboration and the Mediation Structure

The DML analysis provides independent, non-parametric corroboration of the DiD findings and adds a new dimension to the precondition thesis. The Gradient Boosting DML estimate ($hat\{\theta\} = 1,397$) is nearly identical to the DiD $\beta_3 = 1,251$ — remarkable consistency across two methodologically distinct estimators that exploit entirely different sources of variation. The DML additionally quantifies the mediation pathway: when steam vocabulary intensity is controlled, water's direct effect attenuates substantially and loses statistical significance, while steam's independent contribution remains large. This is precisely the footprint one would expect if canals and water mills built the integrated markets and capital base that steam subsequently occupied, rather than the two technologies operating as separate growth drivers.

It bears emphasis that the mediation finding does not diminish the historical importance of water infrastructure — it deepens it. An effect that operates *through* a successor technology is not a small effect; it is a structural precondition. The precondition thesis does not require water to

remain independently significant once steam arrives. It requires only that water came first, built something durable, and that steam's growth potential was bounded by what water had already created. The DML mediation evidence is consistent with all three requirements.

5.6 Implications for the Great Divergence

Consequently, locating water infrastructure as a leading indicator of sustained economic growth encourages a reassessment of the *timeline* of industrialization. The empirical data points toward a distinct sequential transition: not merely the moment humanity transitioned to a mineral economy, but the preceding era where society embedded itself systematically into nature through infrastructural symbiosis — creating the structural preconditions upon which the mineral transition would build.

This empirically reinforces Malm's socio-spatial framing[8]. If a substantial share of the divergence was established in the organic era — our counterfactual analysis suggests 47% by 1810 — then the eventual transition to coal and steam requires more than a strictly thermodynamic explanation. The hydro-social shift provides quantitative evidence that symbiotic scaling was economically viable and temporally prior. The subsequent transition to fossil extraction can thus be viewed as a sociological and structural amplification, operating sequentially after the initial takeoff rather than originating it.

For the Great Divergence debate specifically, our findings suggest that the question "why Britain and not China?" cannot be answered solely by reference to coal deposits or Atlantic trade. Tvedt's (2010) original insight — that Britain's specific hydrological geography enabled a form of water infrastructure that Asian river systems, despite their scale, could not replicate — receives quantitative support from our cross-country DiD. The Asian controls (China, India) show no comparable structural break around 1761, despite possessing extensive river networks and sophisticated hydraulic traditions. The divergence was not merely about water per se, but about the particular institutional and geographical capacity to convert water from a natural endowment into an engineered economic input — the very transformation our NLP analysis captures.

6. Conclusion

This paper provides empirical evidence that refines the technological and macroeconomic timeline of early modernity. By merging unsupervised natural language processing on historical print corpora with formal difference-in-differences economic modeling, we trace the chronological origins of British industrial divergence to an earlier phase than the conventional steam-first narrative suggests.

We find that a measurable structural break in Britain's economic trajectory is closely associated with the 1761 exogenous infrastructure shock of the Bridgewater Canal, validated independently by the 1766 hydro-social linguistic crossover. Britain's post-shock trajectory diverged by ~1,251 per capita relative to its continental peers ($p = 0.042$ with HAC standard errors), with 47% of the ultimate industrial lead established by 1810 — decades before steam power achieved commercial dominance. The treatment effect is economically very large: Cohen's d exceeds 1.2 across all specifications, and the coefficient represents approximately 50% of pre-treatment British GDP per capita. The collapsed DiD estimator of Bertrand et al. (2004) preserves this point estimate across an expanded 13-country panel ($\beta_3 = 1{,}251–1{,}422$), confirming the magnitude's robustness even as the test necessarily lacks statistical power with the cross-sectional units available. Placebo falsification exercises confirm that only the water infrastructure shock uniquely predicts the *timing* of divergence; rival vocabularies (textile, coal, financial) produce noisy or invalid event studies.

Our evidence is consistent with water infrastructure functioning as a necessary precondition rather than a sufficient cause of industrial modernity. The canal era created the integrated markets, accumulated capital, and systemic demand that the subsequent fossil transition would amplify. The channel decomposition supports this sequential reading: when entered simultaneously, the fossil channel dominates the transport channel — a pattern expected if water infrastructure's contribution was absorbed into the broader economic structure it helped create.

Three broader implications follow. First, methodologically, this paper demonstrates that NLP-derived structural breaks can serve as credible mechanism validators when paired with exogenous historical shocks — a technique applicable to other episodes of institutional transformation where discourse shifts precede or accompany material changes. Second, for the Great Divergence debate, our findings suggest that Britain's initial escape from convergence with Asian economies was associated not merely with its access to coal, but with a uniquely favourable hydrological

endowment — gentle river gradients, manageable currents, and consistent rainfall — combined with the institutional capacity to engineer that waterscape into an integrated transport and power network. The contrast with the monsoonal flood regimes and vast, turbulent river systems of China and India underscores that the divergence was not about water per se, but about the specific match between landscape and engineering ambition. Third, for contemporary development economics, the sequential relationship we document between infrastructure-led market integration and subsequent technological amplification may carry lessons for economies whose development strategies must navigate comparable transitions between resource endowments.

Ultimately, Britain's industrial trajectory appears to have been initiated during an era of geographical symbiosis — a period of intense ecological cooperation that established the structural foundations upon which the fossil revolution subsequently built. The empirical data positions hydro-social infrastructure not as an alternative to the steam narrative, but as its necessary and temporally prior precondition.

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7. Limitations and Robustness

The application of a quantitative culturomics framework to historical macroeconomics inherently requires methodological concessions. To ensure transparency, we highlight three primary vulnerabilities in our identification strategy.

7.1 Robustness to Endogenous Treatment Timing

A pervasive vulnerability in applying computational text analysis to macroeconomics is the risk of endogenous treatment timing. If the structural break utilized as the DiD treatment intervention (T_0) is derived exclusively from the linguistic corpus produced by the society experiencing economic transition—e.g., relying solely on our 1766 mathematical NLP crossover —the model risks circularity.

To systematically neutralize this endogeneity critique, our methodology explicitly pivots away from deploying the NLP-derived marker as the treatment baseline. Instead, we anchor the Diff-in-Diff specification to the exogenous historical reality of the 1761 opening of the Bridgewater Canal—a singular infrastructural investment shock universally recognized as the catalyst for the

subsequent canal mania. By utilizing 1761 as an exogenous variable, our use of the 1766 Google Books crossover is elegantly repurposed as an independent NLP mechanism validator. It quantitatively confirms that the exogenous topographical shock of 1761 definitively catalyzed and reorganized the macro-linguistic culture of the society exactly a half-decade later, resolving standard identification concerns cleanly.

7.2 Corpus Bias and the Scientific Publishing Boom

As demonstrated by Pechenick et al. (2015), the Google Books Ngram Corpus does not act as a strictly neutral mirror of popular culture. As the 18th and 19th centuries progressed, the corpus became increasingly dominated by scientific, technical, and legal texts. The surge in vocabulary related to canals and water wheels (`canal`, `water_wheel`, `engineer`) may therefore partially reflect an institutional boom in the publishing of technical manuals and parliamentary navigation acts, rather than a purely linguistic shift in daily life. However, in the context of economic history, the sudden profusion of technical infrastructure literature serves as an incredibly robust proxy for the material reorganization of the underlying economy.

Additionally, our vocabulary selection of 71 semantic targets was fundamentally heuristic. While our placebo vocabulary tournaments demonstrate that the relative timing of the "hydro" crossover uniquely aligns with the onset of GDP divergence compared to plausible alternatives (e.g., textiles, coal, or finance), a purely automated term-clustering algorithm would provide a more systematic foundation for future replication efforts.

7.3 Data Interpolation and Serial Autocorrelation

While the Maddison Project Database (Bolt and van Zanden 2020) provides the finest scale of historical macroeconomic data available, the pre-1800 non-European control groups rely on interpolation between sparse historical benchmarks. To mitigate this concern, our expanded 13-country panel prioritizes control countries with dense pre-treatment coverage: Spain, Portugal, and Poland each contribute 61+ real observations in the pre-1761 period, while France (401), Sweden (401), and Germany (401) provide near-complete annual coverage. China (38 obs.), India (30 obs.), and Japan (27 obs.) contribute to the "Great Divergence" dimension but are interpreted with appropriate caution given their sparser coverage.

Serial autocorrelation is a fundamental challenge for any DiD design on annual time-series GDP data over a 200-year span. The baseline Durbin-Watson statistic of 0.042 confirms extreme positive autocorrelation, consistent with the warnings of Bertrand, Duflo, and Mullainathan (2004). We implement multiple corrections as reported in Table 3:

1. **Newey-West HAC** standard errors (lag=15) inflate the DiD standard error from 208 to 614, but β_3 retains statistical significance at the 5% level ($p = 0.042$).

2. **Country-clustered standard errors** yield $p < 0.001$, though with only $G = 3$ European clusters, clustered inference should be interpreted conservatively following Cameron, Gelbach, and Miller (2008).

3. **The collapsed DiD estimator** of Bertrand et al. (2004) completely eliminates serial autocorrelation by construction. We expand the control group to 13 countries to maximize statistical power. The point estimate is remarkably stable across panel sizes ($\beta_3 = 1{,}251$–$1{,}422$), confirming that the estimated magnitude is not an artifact of autocorrelation-inflated precision. With the full 13-country panel ($N = 26$), the collapsed p-value is 0.280 — an improvement over the 3-country panel ($p = 0.628$) but still below conventional significance thresholds. This reflects a structural power limitation inherent to collapsed estimation in cross-country macro designs where treatment is genuinely nation-specific.

4. **Magnitude metrics** provide an alternative lens for assessing the treatment effect. Cohen's d exceeds 1.2 across all collapsed specifications, indicating a very large effect size. The treatment effect represents approximately 49–55% of pre-treatment British GDP per capita — an economically enormous divergence that is substantively meaningful regardless of the collapsed estimator's p-value.

Taken together, these results indicate that the *magnitude* and *economic significance* of the treatment effect are robust across all corrections and panel sizes. The statistical precision of the collapsed estimator improves monotonically with panel size, suggesting that the non-significance reflects insufficient cross-sectional units rather than absence of a real effect. The HAC correction, which preserves the full time-series information while accounting for serial dependence, represents the most informative inferential framework for this setting.

7.4 DML Limitations: Single Treated Unit and Placebo Validity

The DML estimator introduces its own identification constraints that require transparent disclosure.

Single treated unit. All treatment variation in the DML specification originates from a single country (Britain). The cluster-robust standard errors correctly account for this by grouping influence functions at the country level, but the fundamental power limitation of $G = 1$ treated cluster remains. The Neyman-orthogonal SE reported alongside the cluster-robust SE should be interpreted as a lower bound on uncertainty, not a preferred estimate. The consistency between the DML $\hat{\theta} \approx 1{,}397$ and the DiD $\beta_3 = 1{,}251$ across these two methods — each with different identifying assumptions — provides some reassurance that the estimate is not purely an artifact of any single inferential framework.

ML method sensitivity. Linear methods (Lasso, Ridge) produce inflated estimates ($\hat{\theta} \approx 7{,}000$) because they cannot flexibly absorb the nonlinear year-vocabulary trend; the vocabulary intensity series is highly nonlinear in time, and a quadratic polynomial is an imperfect control. Random Forest produces a degenerate cluster-robust SE in some specifications because the tree learner fits the year-trend well enough to nearly collapse $\hat{\varepsilon}_D \approx 0$, making the sandwich estimator numerically unstable. These are well-known failure modes of DML under near-perfect treatment predictability (Chernozhukov et al. 2018, Remark 3.2). Gradient Boosting, which balances flexibility with regularization, is the preferred specification and produces estimates consistent with the DiD.

Geographic placebo test. The in-space DML placebo — assigning treatment to each control country in turn — does not produce uniformly null results for European controls. France ($\hat{\theta} = 3{,}265$), the Netherlands ($\hat{\theta} = 4{,}014$), and Germany ($\hat{\theta} = 5{,}763$) all show positive significant pseudo-effects. This is expected on reflection: the treatment intensity variable is the *English-language* vocabulary index, a time series common to all specifications. Any country whose GDP rose over the 1700–1900 period will mechanically correlate with this monotonically trending series, regardless of whether water infrastructure played a role in its economy. The geographic placebo is therefore uninformative in this context. The more valid falsification tests are the *vocabulary tournament* (only water, not coal, textiles, or finance, produces a clean event study aligned with 1761) and the *temporal ordering* (water vocabulary crossover in 1766 precedes the steam transition by four decades). Asian controls — China, India, Japan — do show the

correct negative pseudo-effect ($\hat{\theta} < 0$), consistent with non-industrialization during this period and with the Great Divergence framework. We report these results in full rather than selectively, and caution against over-interpreting the geographic placebo as either supporting or refuting the water infrastructure thesis.

Data Availability Statement

All data and code required to reproduce the analyses in this paper are publicly available at [REPOSITORY URL REDACTED FOR REVIEW]. Historical GDP per capita data are sourced from the Maddison Project Database 2023 (Bolt and van Zanden 2020), accessible via the Dataverse repository. Linguistic frequency data are drawn from the Google Books Ngram Corpus (`eng_gb_2019`), publicly available at https://books.google.com/ngrams. A self-contained replication package including all scripts, data files, and output figures is available as a supplementary archive.

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